

Lecture 11: Panel data I

PPHA 34600
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From last time: finishing IV

Z_i is a valid instrument when the following are satisfied:

- 1 **First stage:** $\text{Cov}(Z_i, D_i) \neq 0$
- 2 **Exclusion restriction:** $\text{Cov}(Z_i, \varepsilon_i) = 0$

When we have these two conditions, we can...:

- Estimate causal effects
- ...but only for compliers!

Further down the SOU rabbit hole

So far, we've focused on data across units. Now we'll add time:

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 - What we've been using
 - Observations across units at a single point in time

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- **Repeated cross-section data:**
 - Repeated sampling of different units over time
- **Panel data:**
 - Multiple observations of the same unit over time

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 - Getting a clean comparison means separating τ from all of these differences
- Enter time series data:
 - **Fundamental insight:**
Rather than comparing i to j , compare i in t to i in $t - 1$
 - In this formulation, i serves as a control for itself
 - i am much more similar to myself yesterday than i am to j

Making time-series comparisons

Consider a setting with only one unit:

- We now denote our outcome as $Y_t(D_t)$ (no subscript: only one unit)
- As usual, we want to estimate $\tau^{ATE} = E[Y_t(D_t = 1) - Y_t(D_t = 0)]$
- But we can't observe both $Y_{t=1}(D_{t=1} = 1)$ and $Y_{t=1}(D_{t=1} = 0)$
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- Suppose in $t = 0$, $D_{t=0} = 0$, and in $t = 1$, $D_{t=1} = 1$
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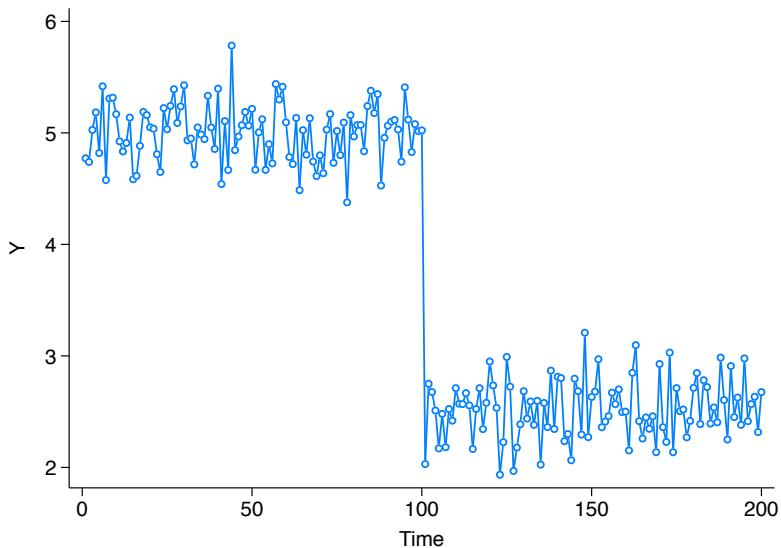
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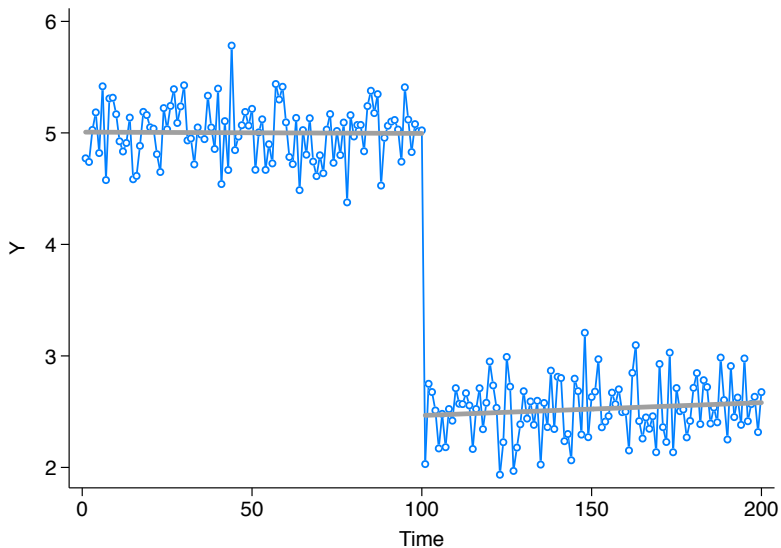
- We can also extend this to many periods:

$$\hat{\tau}^{TS} = \bar{Y}_{t \in \text{post}} - \bar{Y}_{t \in \text{pre}}$$

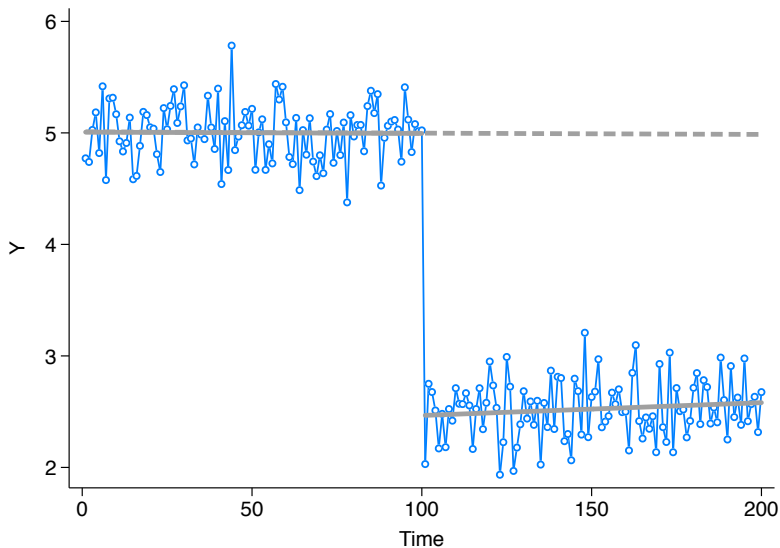
Time series, visually



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This time series approach compares unit i to itself over time:

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- Enter the difference estimator:

$$\begin{aligned} Y_{i,t=1} - Y_{i,t=0} &= \tau(D_{i,t=1} - D_{i,t=0}) + \beta(X_i - X_i) \\ &= \tau(D_{i,t=1} - D_{i,t=0}) \\ &= \tau(1 - 0) \\ &= \tau \end{aligned}$$

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Time series: identifying assumptions

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to recover the true τ , we need an important assumption.

Consider the DGP:

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→ **Any time-varying variables will create bias in $\hat{\tau}^{TS}$**

→ Also true for observables: we can't separate D_{it} from coincident V_{it}

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Another way to think about this assumption:

- Y_i would be unchanged in the absence of treatment
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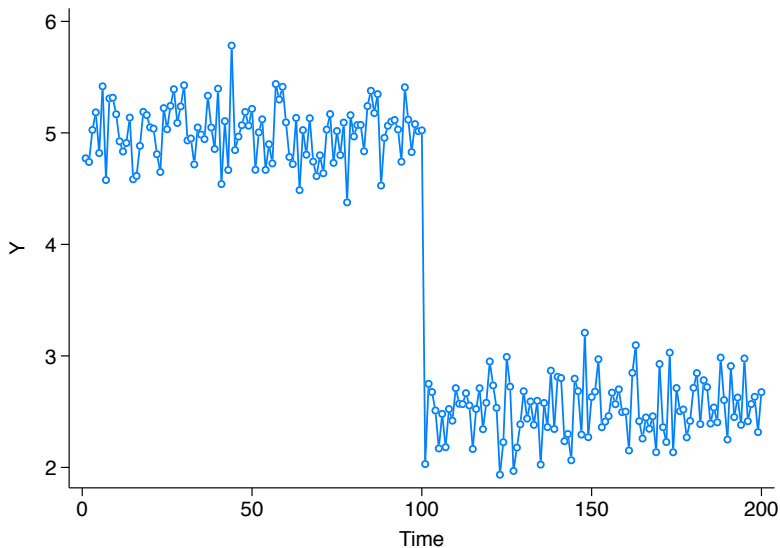
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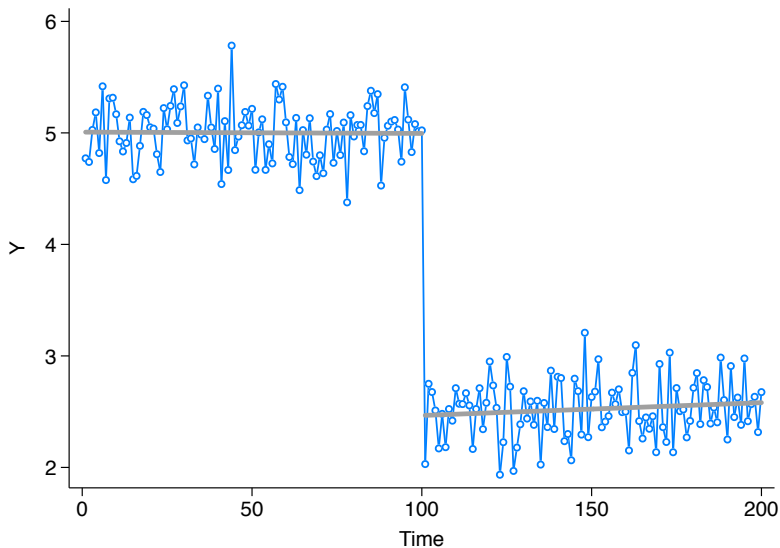
$$= \tau + \text{counterfactual trend}$$

We have to assume that the counterfactual trend is zero

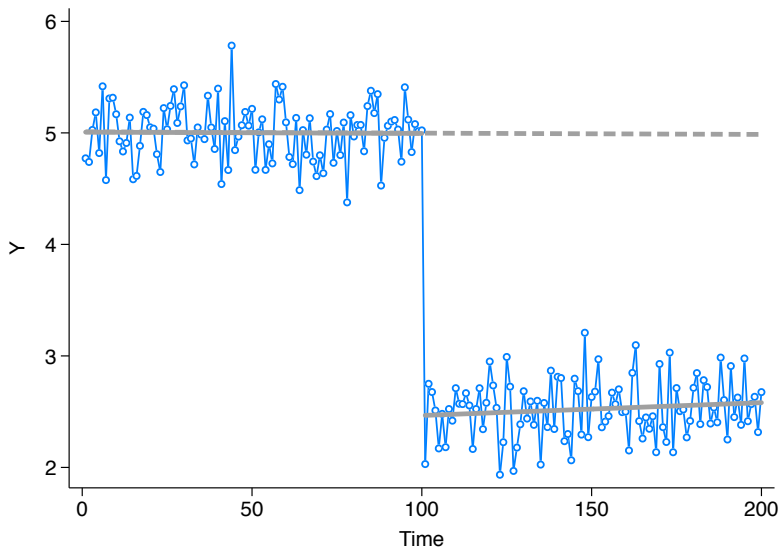
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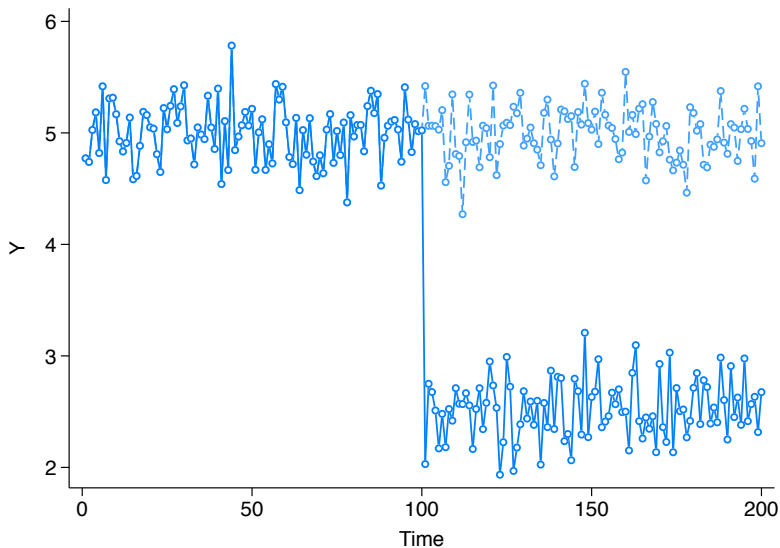
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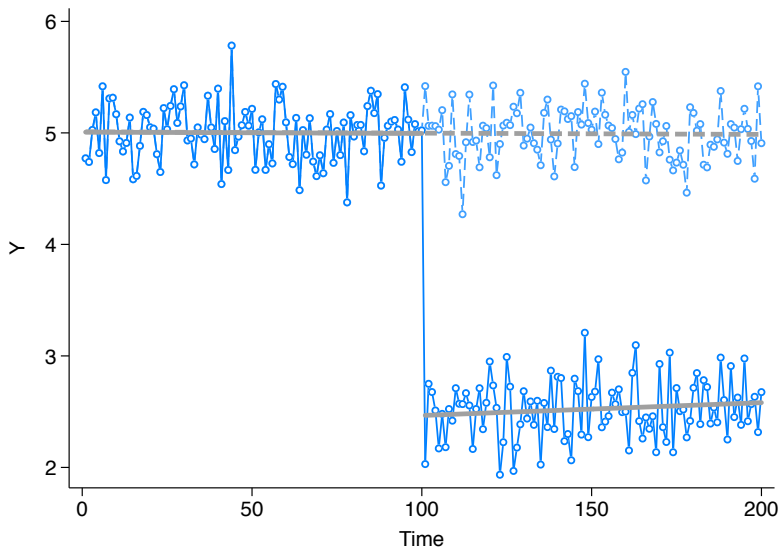
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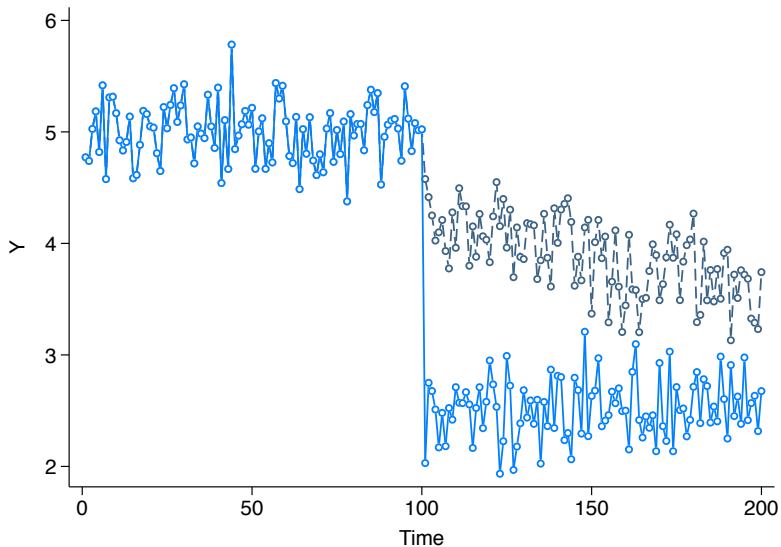
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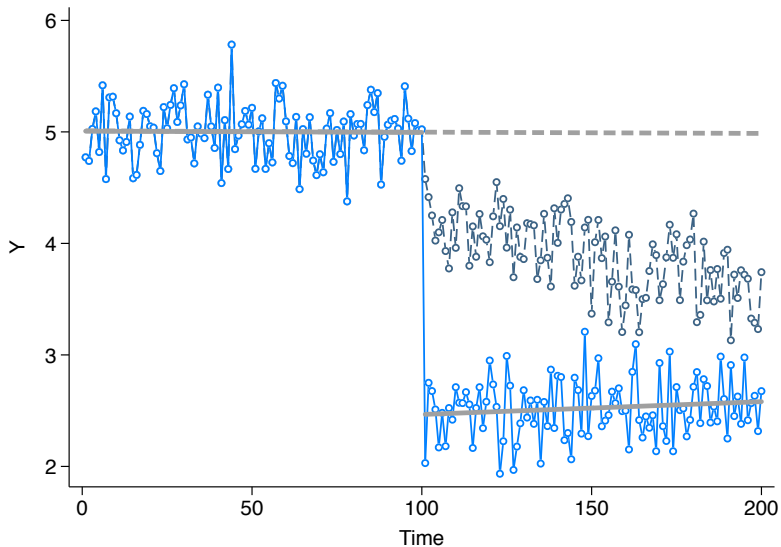
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Why does this matter?

- If the assumption is not satisfied, we will confound the trend with τ
- We cannot eliminate the trend with one time series

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Using panel data, we can combine two bad estimators into a good one:

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- We can combine these into the **difference-in-differences** estimator:
 - Uses across-unit, within time, comparisons
 - And within-unit, across time, comparisons

Differences-in-differences (DD)

The problem with time series is the counterfactual trend:

- How would treated i have behaved in $t = 1$ without treatment?
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This compares **treated** to **untreated** units **over time**

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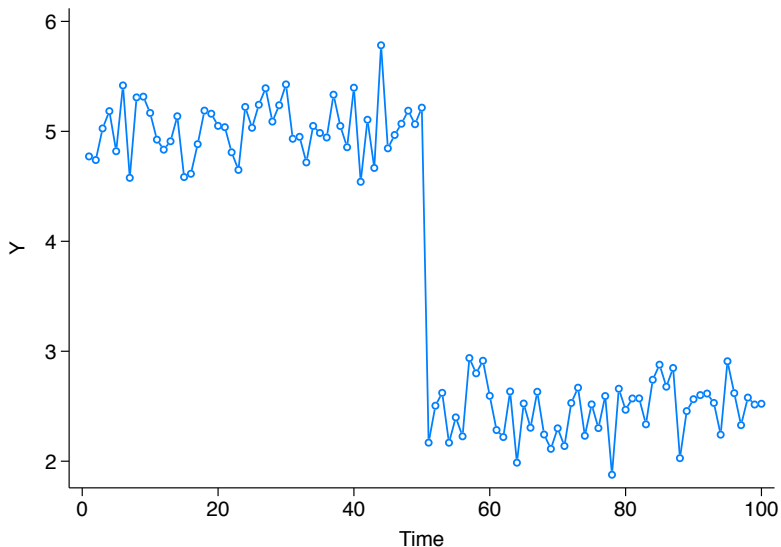
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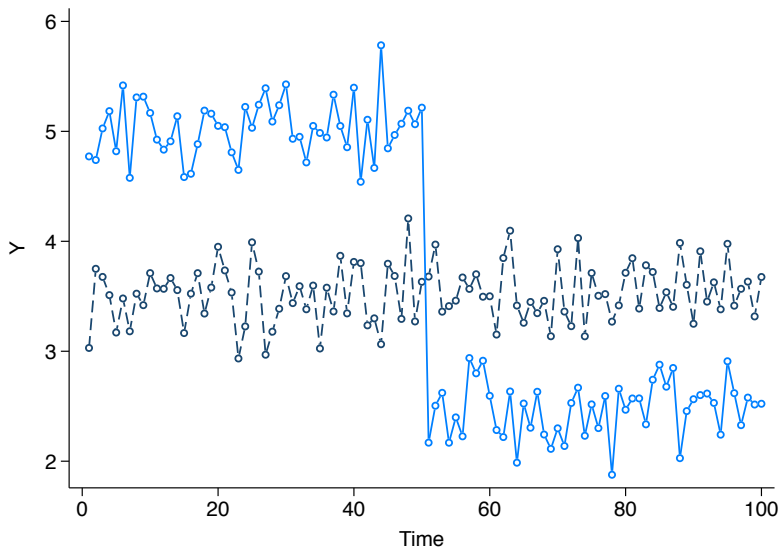
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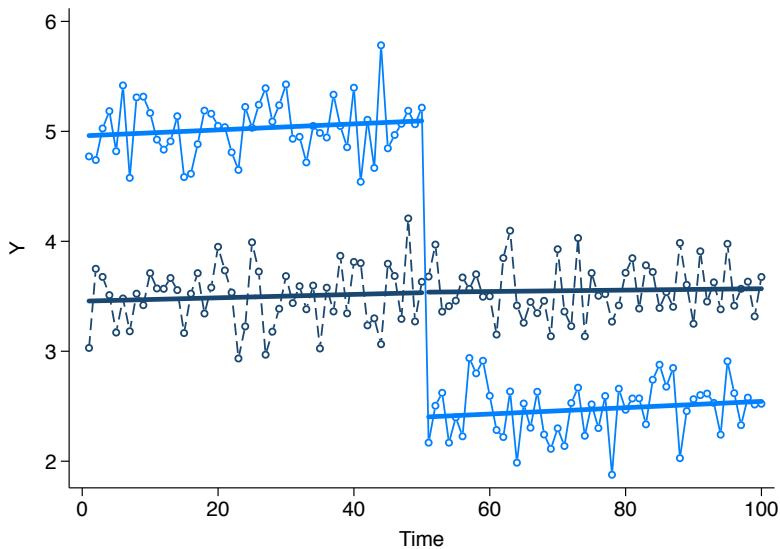
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The identifying assumption of the DD is “parallel trends”:

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$$\begin{aligned}\hat{\tau}^{DD} &= (Y_{i,t=1}(D_{it} = 1) - Y_{i,t=0}(D_{it} = 0)) - (Y_{j,t=1}(D_{jt} = 0) - Y_{j,t=0}(D_{jt} = 0)) \\ &= (Y_{i,t=1}(1) - Y_{i,t=0}(0)) - (Y_{j,t=1}(0) - Y_{j,t=0}(0)) \\ &\quad + (Y_{i,t=1}(0) - Y_{i,t=1}(0)) \\ &= (Y_{i,t=1}(1) - Y_{i,t=1}(0)) + [(Y_{i,t=1}(0) - Y_{i,t=0}(0)) - (Y_{j,t=1}(0) - Y_{j,t=0}(0))] \\ &\approx E[Y_{i,t=1}(1) - Y_{i,t=1}(0)|D_i = 1] + E[Y_{i,t=1}(0) - Y_{i,t=1}(0)|D_i = 1] \\ &\quad - E[Y_{j,t=1}(0) - Y_{j,t=0}(0)|D_j = 0]\end{aligned}$$

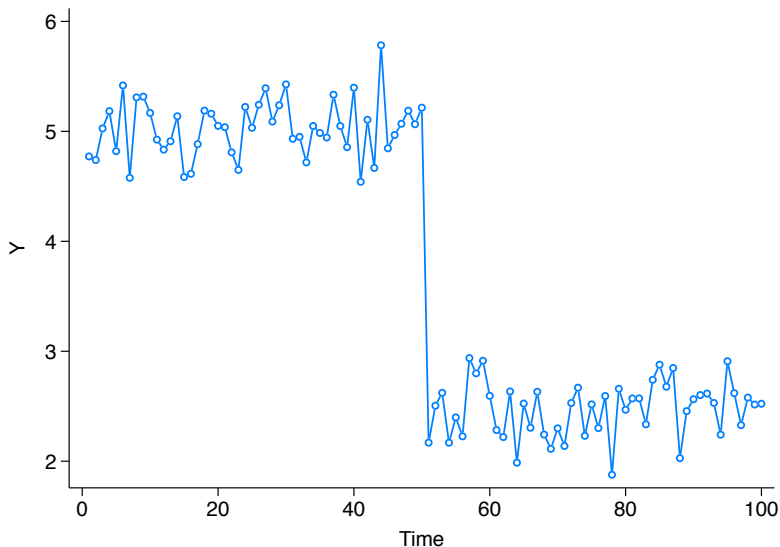
In other notation...

The identifying assumption of the DD is “parallel trends”:

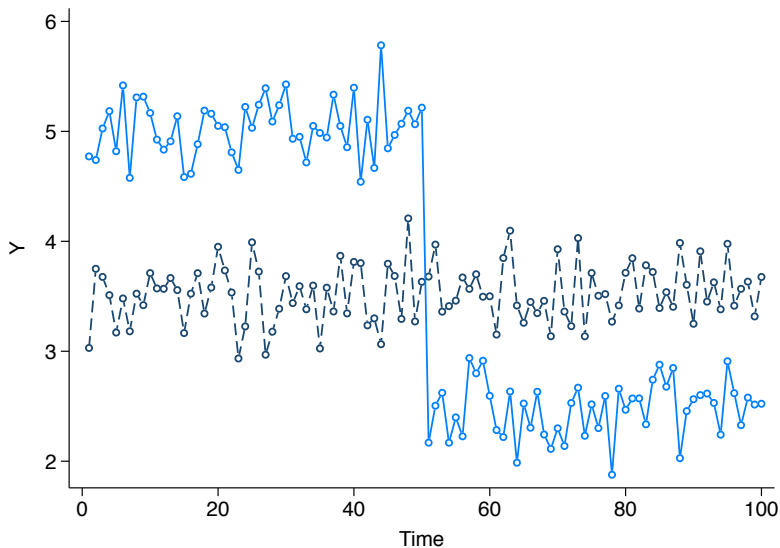
$$\begin{aligned}\hat{\tau}^{DD} &= (Y_{i,t=1}(D_{it} = 1) - Y_{i,t=0}(D_{it} = 0)) - (Y_{j,t=1}(D_{jt} = 0) - Y_{j,t=0}(D_{jt} = 0)) \\&= (Y_{i,t=1}(1) - Y_{i,t=0}(0)) - (Y_{j,t=1}(0) - Y_{j,t=0}(0)) \\&\quad + (Y_{i,t=1}(0) - Y_{i,t=1}(0)) \\&= (Y_{i,t=1}(1) - Y_{i,t=1}(0)) + [(Y_{i,t=1}(0) - Y_{i,t=0}(0)) - (Y_{j,t=1}(0) - Y_{j,t=0}(0))] \\&\approx E[Y_{i,t=1}(1) - Y_{i,t=1}(0) | D_i = 1] + E[Y_{i,t=1}(0) - Y_{i,t=1}(0) | D_i = 1] \\&\quad - E[Y_{j,t=1}(0) - Y_{j,t=0}(0) | D_j = 0] \\&= \tau + \text{counterfactual trend} - \text{untreated trend}\end{aligned}$$

Identifying assumption: untreated trend = counterfactual trend

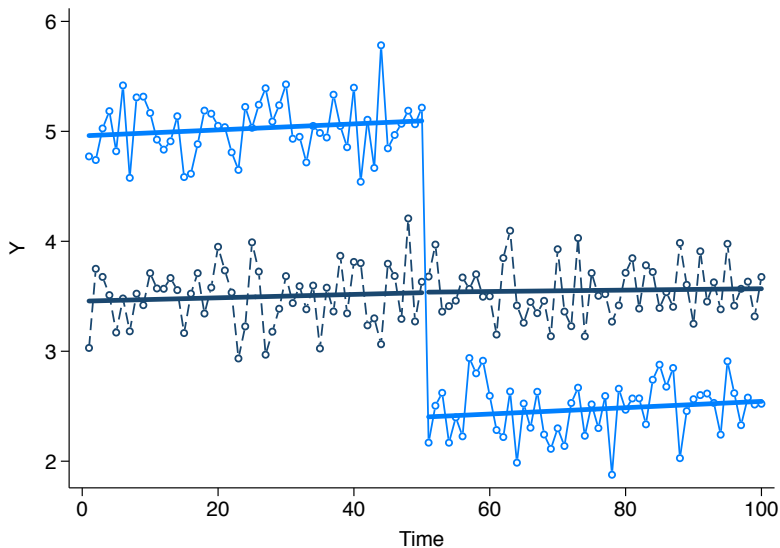
Identifying assumptions, visually



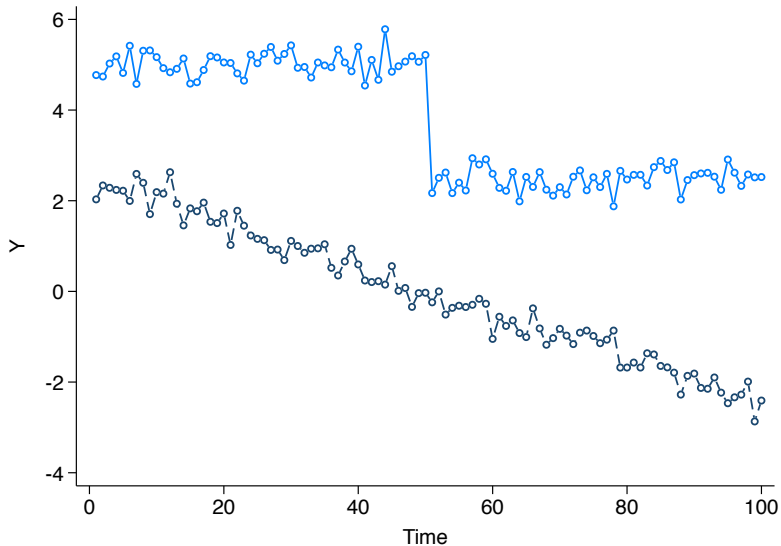
Identifying assumptions, visually



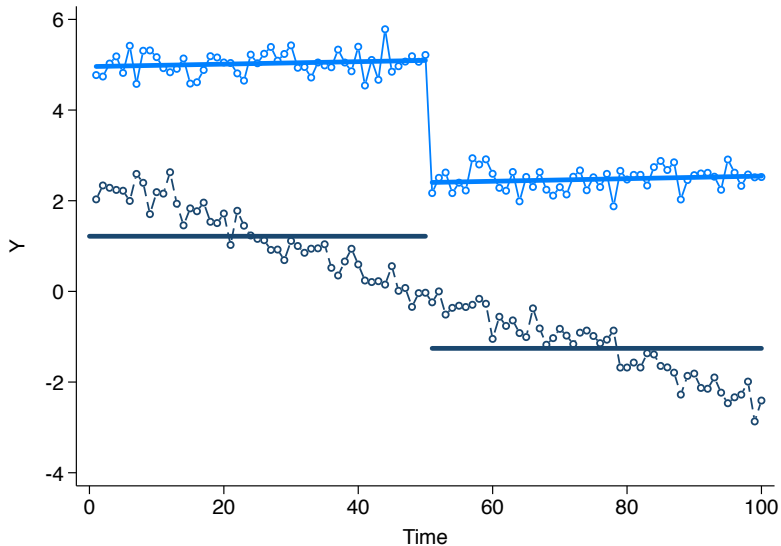
Identifying assumptions, visually



Identifying assumptions, visually



Identifying assumptions, visually



Implementing DD via regression

The simplest implementation of DD is just:

$$\hat{\tau}^{DD} = (\bar{Y}(\text{treat}, \text{post}) - \bar{Y}(\text{treat}, \text{pre})) - (\bar{Y}(\text{untreat}, \text{post}) - \bar{Y}(\text{untreat}, \text{pre}))$$

Implementing DD via regression

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We can implement this via the following regression:

$$Y_i = \alpha + \tau \text{Treat} \times \text{Post}_{it} + \beta \text{Treat}_i + \delta \text{Post}_t + \varepsilon_{it}$$

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To link these together, see:

$$\bar{Y}(\text{treat}, \text{post}) = \hat{\alpha} + \hat{\tau} + \hat{\beta} + \hat{\delta}$$

$$\bar{Y}(\text{treat}, \text{pre}) = \hat{\alpha} + \hat{\beta}$$

$$\rightarrow \bar{Y}(\text{treat}, \text{post}) - \bar{Y}(\text{treat}, \text{pre}) = \hat{\delta} + \hat{\tau}$$

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$$\rightarrow \bar{Y}(\text{treat}, \text{post}) - \bar{Y}(\text{treat}, \text{pre}) = \hat{\delta} + \hat{\tau}$$

and

$$\bar{Y}(\text{untreat}, \text{post}) = \hat{\alpha} + \hat{\delta}$$

$$\bar{Y}(\text{untreat}, \text{pre}) = \hat{\alpha}$$

$$\rightarrow \bar{Y}(\text{untreat}, \text{post}) - \bar{Y}(\text{untreat}, \text{pre}) = \hat{\delta}$$

Implementing DD via regression

The simplest implementation of DD is just:

$$\hat{\tau}^{DD} = (\bar{Y}(\text{treat}, \text{post}) - \bar{Y}(\text{treat}, \text{pre})) - (\bar{Y}(\text{untreat}, \text{post}) - \bar{Y}(\text{untreat}, \text{pre}))$$

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$$\bar{Y}(\text{treat}, \text{post}) = \hat{\alpha} + \hat{\tau} + \hat{\beta} + \hat{\delta}$$

$$\bar{Y}(\text{treat}, \text{pre}) = \hat{\alpha} + \hat{\beta}$$

$$\rightarrow \bar{Y}(\text{treat}, \text{post}) - \bar{Y}(\text{treat}, \text{pre}) = \hat{\delta} + \hat{\tau}$$

and

$$\bar{Y}(\text{untreat}, \text{post}) = \hat{\alpha} + \hat{\delta}$$

$$\bar{Y}(\text{untreat}, \text{pre}) = \hat{\alpha}$$

$$\rightarrow \bar{Y}(\text{untreat}, \text{post}) - \bar{Y}(\text{untreat}, \text{pre}) = \hat{\delta}$$

Running this regression yields $\hat{\tau} = \hat{\tau}^{DD}$

A handy table

We can implement DD via the following regression:

$$Y_i = \alpha + \tau Treat \times Post_{it} + \beta Treat_i + \delta Post_t + \varepsilon_{it}$$

A handy table

We can implement DD via the following regression:

$$Y_i = \alpha + \tau \textit{Treat} \times \textit{Post}_{it} + \beta \textit{Treat}_i + \delta \textit{Post}_t + \varepsilon_{it}$$

This gives us:

	Pre	Post	Difference
Treated	$\alpha + \beta + \delta + \tau$	$\alpha + \beta$	$\delta + \tau$
Untreated	$\alpha + \delta$	α	δ
Difference	$\beta + \tau$	β	τ

Adding covariates

We can add covariates:

$$Y_i = \alpha + \tau \textit{Treat} \times \textit{Post}_{it} + \beta \textit{Treat}_i + \delta \textit{Post}_t + \gamma X_{it} + \varepsilon_{it}$$

Adding covariates

We can add covariates:

$$Y_i = \alpha + \tau Treat \times Post_{it} + \beta Treat_i + \delta Post_t + \gamma X_{it} + \varepsilon_{it}$$

We do this to:

- ① Add precision: like the RCT, we can soak up variation
- ② Control for important observables
 - This mixes DD with SOO

An example: Mobile phones and markets

Policy issue:

- Imperfect information can complicate markets
- Can cell phones help?

Approach:

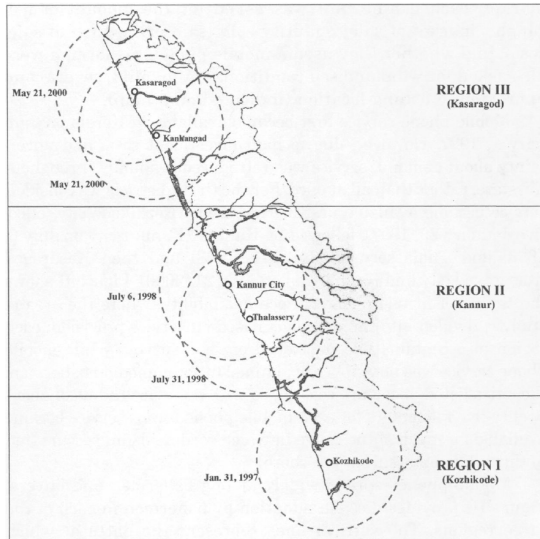
- Mobile phone service came to Kerala between 1997 and 2001
- We want to know the effect of phones on welfare
- Nobody ran an RCT to roll phones out...
- ...but phones got to different locations at different times

Motivating evidence

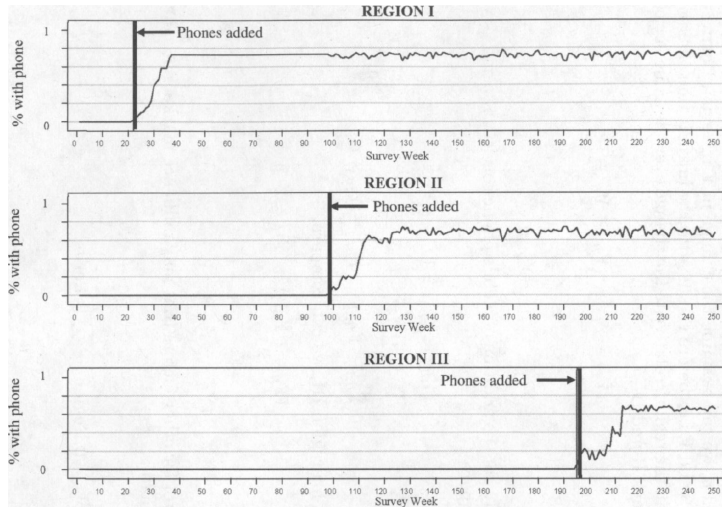
PRICES AND EXCESS SUPPLY AND DEMAND IN FIFTEEN SARDINE BEACH MARKETS

	Price (Rs/kg)	Excess buyers	Excess sellers
Kasaragod District			
Hosabethe	6.2	0	0
Aarikkadi	4.0	0	0
Kasaba	0.0	0	4
Kanhangad	7.2	0	0
Thaikadappuram	9.7	11	0
Kannur District			
Puthiangadi	8.7	2	0
Neerkkadavu	6.9	0	0
Ayikkara	8.4	1	0
Thalassery	4.3	0	0
New Mahe	6.2	0	0
Kozhikode District			
Chombala	9.9	15	0
Badagara	0.0	0	11
Quilandi	9.8	12	0
Puthiyangadi	0.0	0	6
Chaliyam	6.4	0	0

The natural experiment



The natural experiment



Estimating the effects of mobile phones on welfare

The authors will run a (simplified) version of:

$$\hat{\tau} = (\bar{Y}(treat, post) - \bar{Y}(treat, pre)) - (\bar{Y}(untreat, post) - \bar{Y}(untreat, pre))$$

Where:

$\bar{Y}$ is the average of the outcome

- (He'll actually do this for regions I, II, and III separately)

	Period 0 (pre-phone)
Percent of fishermen who fish in local catchment zone	
Region I	0.98 (0.003)
Region II	0.99 (0.002)
Region III	0.98 (0.002)
Percent of fishermen who sell in local catchment zone	
Region I	1.00 (0.00)
Region II	1.00 (0.00)
Region III	1.00 (0.00)
Number of fishing units	
Region I	83
Region II	69
Region III	53

Balance

	Period 0 (pre-phone)
Max-min spread (Rs/kg)	
Region I	7.60 (0.50)
Region II	8.19 (0.44)
Region III	8.24 (0.47)
Coefficient of variation (percent)	
Region I	.68 (0.07)
Region II	.62 (0.04)
Region III	.69 (0.09)
Waste (percent)	
Region I	0.08 (0.01)
Region II	0.05 (0.01)
Region III	0.07 (0.01)

Supporting evidence: Prices

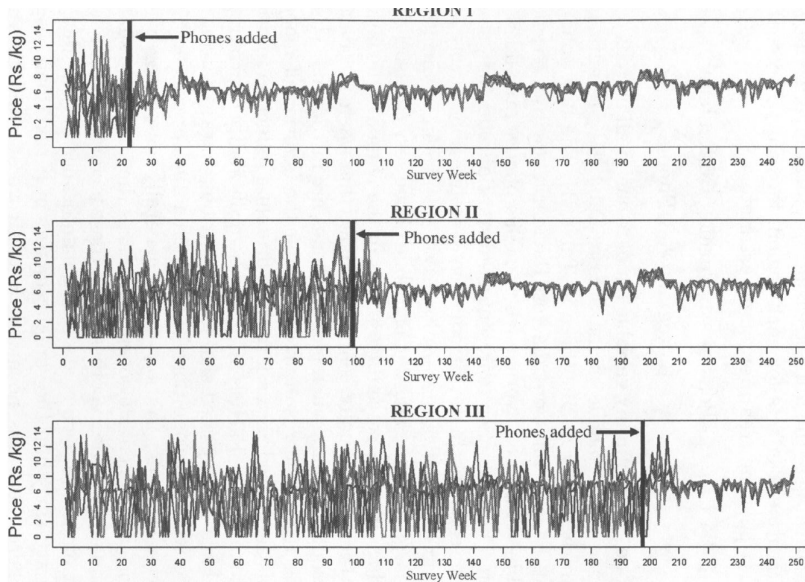


TABLE V
ESTIMATED EFFECTS OF MOBILE PHONES ON MARKET OUTCOMES:
SEPARATE TREATMENTS

	Max-min spread	Coefficient of variation	Waste
Estimated effects of adding phones to region I			
(a) Using region II as the control group	-4.8	-.46	-0.064
$(Y_{I,1} - Y_{I,0}) - (Y_{II,1} - Y_{II,0}) = \beta_{RI_P1}$ - β_{RII_P1}	(0.68)	(0.07)	(0.005)
(b) Using region III as the control group	-4.8	-.42	-0.060
$(Y_{I,1} - Y_{I,0}) - (Y_{III,1} - Y_{III,0}) = \beta_{RI_P1}$	(0.68)	(0.07)	(0.005)
Estimated effects of adding phones to region II			
(c) Using region I as the control group	-5.8	-.39	-0.039
$(Y_{II,2} - Y_{I,1}) - (Y_{I,2} - Y_{I,1}) = \beta_{RII_P2}$ - $\beta_{RII_P1} - \beta_{RI_P2} + \beta_{RI_P1}$	(0.43)	(0.05)	(0.003)
(d) Using region III as the control group	-4.9	-.36	-0.038
$(Y_{II,2} - Y_{II,1}) - (Y_{III,2} - Y_{III,1}) = \beta_{RII_P2}$ - β_{RII_P1}	(0.43)	(0.05)	(0.003)
Estimated effects of adding phones to region III			
(e) Using region I as the control group	-4.9	-.38	-0.055
$(Y_{III,3} - Y_{III,2}) - (Y_{I,3} - Y_{I,2}) = \beta_{RI_P2}$ - β_{RI_P3}	(0.48)	(0.05)	(0.004)
(f) Using region II as the control group	-4.7	-.35	-0.054
$(Y_{III,3} - Y_{III,2}) - (Y_{II,3} - Y_{II,2}) = \beta_{RII_P2}$ - β_{RII_P3}	(0.48)	(0.05)	(0.004)

VIOLATIONS OF THE LAW OF ONE PRICE

	Period 0 (pre-phone)	Period 1 (region I has phones)	Period 2 (region II has phones)	Period 3 (region III has phones)
Overall				
Region I	0.54	0.03	0.04	0.03
Region II	0.57	0.55	0.06	0.05
Region III	0.60	0.58	0.58	0.08
With time + depreciation				
Region I	0.50	0.01	0.02	0.02
Region II	0.53	0.52	0.03	0.03
Region III	0.57	0.55	0.54	0.05
All markets combined				
Without time + depreciation	0.47	0.35	0.20	0.05
With time + depreciation	0.44	0.31	0.16	0.03

Results

TABLE 11
EFFECTS OF MOBILE PHONES ON PRODUCERS AND CONSUMERS: POOLED TREATMENTS

	(1) Quantity sold	(2) Price	(3) Price (if >0)	(4) Revenue	(5) Costs	(6) Profits	(7) Profit users	(8) Profit nonuser	(9) Consumer price	(10) Consumer surplus
Phone	23 (8.4)	-.05 (0.03)	-.44 (0.03)	205 (62)	72 (5.6)	133 (60)	184 (90)	97 (47)	-.39 (0.22)	.14 (0.04)
Region I	36 (6.6)	.25 (0.03)	-.19 (0.03)	370 (56)	3.7 (4.9)	367 (54)	458 (77)	306 (44)	.51 (.30)	-.11 (0.03)
Region II	22 (5.2)	.03 (0.02)	-.07 (0.02)	173 (42)	3.3 (3.0)	170 (40)	204 (57)	130 (35)	.38 (0.27)	-.03 (0.02)
Period 1	-5.3 (10)	.48 (0.03)	.36 (0.03)	66 (59)	7.6 (4.2)	58 (58)	63 (94)	61 (43)	.22 (0.05)	-.16 (0.04)
Period 2	-17 (14)	.64 (0.04)	.51 (0.03)	34 (80)	2.3 (3.7)	32 (80)	-6.3 (122)	62 (57)	.65 (0.27)	-.30 (0.05)
Period 3	-7.8 (16)	1.0 (0.05)	.84 (0.04)	215 (99)	16 (6.0)	200 (97)	212 (145)	189 (74)	.81 (0.35)	-.48 (0.05)
Observations	74,700	74,700	73,335	74,700	74,700	74,700	41,012	33,688	3,735	3,735

TL;DR:

- ① We can leverage time series data for identification
- ② This is more powerful when combined with cross-section
- ③ The resulting diff-in-diff is one of the better quasi-experiments

For next class

Topics:

- Panel data II

Reading: Hsiang and Jina (2014). You can skip:

- Section 2
- pp 32– 49 (start again with Summary and Discussion)